MATH70113 Simulation Methods for Finance 2025-26
Group Assignment

1 Guidelines

1.1 General information

- **Group structure**: Each group should consist of 3-4 members. Please form your own group.
- **Marking**: This assignment accounts for 20% of your module grade. All members within the same group will be awarded the same mark. This mark is determined by the collective performance of submitted work, taking into account factors such as the extent of exploration, the Python program, and the quality of the written report.

1.2 Submission

The coursework must be submitted by

Friday 10 April 2026, 2pm (UK time).

- Consequences regarding late submission can be found on the College's website here.

  Note in particular: Work submitted up to 24 hours after the assessment deadline will be marked but capped at the passmark. Work submitted more than 24 hours late will not be accepted as a valid attempt and mark of zero will be recorded.

- A representative of each group should upload all necessary files **in a single zip-file** to the coursework dropbox on Blackboard. Please name the file using the surnames of all group members in form of "Surname1_Surname2_Surname3_Surname4.zip".

- The zip-file should contain the following:

  - One **Jupyter notebook file (.ipynb)**, that the examiners can run, and that contains the python code used to generate the numerical results to all questions. Write your code with necessary comments.
  - One **pdf file** contains a written report that covers problem description, solution methodology, computational results, statistical analysis, graphs, etc. The pdf file (excluding the cover page) should not exceed 15 pages of A4 paper with minimum font size 11pt. One may include additional materials in appendix beyond 15 pages but that will not be assessed. **The cover page of the report should clearly display the names and CIDs of all group members**.

- Please write your report using LaTeXand write your code using Python.

1.3 Questions of the coursework

The coursework comprises two components:

- Part 1 encompasses fundamental tasks that all groups are required to complete.
- Part 2 involves the review of elective papers. Each group can choose freely which paper to work on. The report should summarise the problem, the solution methodology, reproduce numerical results, and potentially explore relevant literature. It is crucial to note that merely reiterating theoretical results is insufficient. Instead, you are expected to validate these theoretical results through numerical experiments.

  Additional bonus points will be awarded for comprehensive and in-depth exploration of the subject, including but not limited to conducting numerical tests beyond those presented in the paper, comparing the paper with other related literature, and engaging in a discussion on the pros and cons of the proposed method. The overall direction and expectation are to encourage thorough and extensive studies of the subject matter.

Please provide a very brief breakdown of tasks/duties performed by each member at the end of the written report, even though every member in the group will receive the same mark. This (short) section will not be counted towards the 15-page limit of the report.

2 Part 1

If you use a random number generator in any of the following questions, please seed your generator so that the results are reproducible. Justify your simulation procedure with necessary mathematical derivations and proper visualisations.

Problem 1. *Suppose that S satisfies the standard Geometric Brownian Motion:*

$$dS_t = rS_t dt + \sigma S_t dW_t,\ t \in [0, T]; \quad S_0 = s_0.$$

Consider a down-and-out barrier option with the discounted payoff:

$$V = \mathbb{E}\left[e^{-rT}(S_T - K)^+ 1_{\min_{t\in[0,T]} S_t > B}\right].$$

Take the parameter values $r = 0.05$, $\sigma = 0.5$, $T = 1$, $s_0 = 100$, $K = 110$ and $B = 90$.

Implement the pathwise sensitivity method to estimate Delta and Vega for the down-and-out barrier option, incorporating the barrier crossing technique introduced in lectures to achieve $\mathcal{O}(h)$ weak convergence. For additional guidance on the barrier crossing technique, you can refer to Section 6.4 of the textbook by Glasserman [3].

3 Part 2

Apart from the basic task in Part 1, each group shall complete one task below. You can choose freely which one to work on. You are encouraged to do broad and deep studies for each specific task.

Problem 2. *Conduct a review of the paper titled "Multilevel Monte Carlo path simulation," authored by Giles [2]. This paper introduces a novel variance reduction technique designed to reduce the computational complexity associated with estimating expected values arising from stochastic differential equations through Monte Carlo path simulation. Please explore related papers on various other applications of multilevel Monte Carlo methods.*

Problem 3. *Conduct a review of the paper "Number of paths versus number of basis functions in American option pricing," authored by Glasserman and Yu [4]. This paper provides theoretical guidance on the Longstaff-Schwartz method proposed in [5] for American option pricing. In particular, it addresses the selection of the regression basis and the determination of the number of simulated paths.*

Problem 4. *Conduct a review of the paper "Deep optimal stopping," authored by Becker, Cheridito, and Jentzen [1]. This paper introduces a deep learning method for optimal stopping problems, which directly learns the optimal stopping rule from Monte Carlo samples. To assess its efficacy, you have to compare the performance of the deep learning method with the Longstaff-Schwartz method in [5] for suitable one-dimensional problems. Moreover, you are expected to examine the algorithm's scalability in a multi-dimensional setting.*

References

[1] Sebastian Becker, Patrick Cheridito, and Arnulf Jentzen. Deep optimal stopping. *The Journal of Machine Learning Research*, 20(1):2712–2736, 2019.

[2] Michael B Giles. Multilevel monte carlo path simulation. *Operations research*, 56(3):607–617, 2008.

[3] Paul Glasserman. *Monte Carlo methods in financial engineering*, volume 53. Springer, 2004.

[4] Paul Glasserman and Bin Yu. Number of paths versus number of basis functions in american option pricing. *Ann. Appl. Probab.*, 14(1):2090–2119, 2004.

[5] Francis A Longstaff and Eduardo S Schwartz. Valuing american options by simulation: a simple least-squares approach. *The review of financial studies*, 14(1):113–147, 2001.